the problem

Users losing long streaks experience genuine emotional distress that can drive permanent churn.

These should be Duolingo's most valuable users, with proven commitment, high lifetime value, and strong subscription potential.
But streak loss can turn Duolingo's most powerful engagement mechanism into a quit trigger.

"I got busy and lost my 2 year streak a couple weeks ago. Guess it's a sign to delete as well 🤦‍♀️"
TikTok User

"I think I was at 185 days when I forgot for a weekend. You're not getting me to start over. It's not happening."
My friend who refuses to start a friend streak with me

Hypothesis

When users lose long streaks, they churn at higher rates because they lose motivation to return.

A forgiving mechanism that motivates users to re-engage with core learning to retain their streak would transform a demotivation moment into renewed learning.

Primary KPI: iWAURR

This mechanism would aim to improve iWAURR by reactivating users in the critical window after a streak is broken and before they fall dormant.

With nearly 8M users with streaks over a year (2024), improving iWAURR for this high-value segment would then feed into protecting the topline metric of DAU.

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the solution

The Phoenix Challenge is a feature to turn streak loss from a dead end to a comeback opportunity.

Upon losing a 100+ day streak and expending all freezes, users can opt into a challenge: completing the next 3 path lessons within 72 hours to fully restore their original streak. Free users will be eligible for this challenge once every 365 days, while Super and Max subscribers will be eligible monthly.
The intention is to leverage loss aversion to drive re-engagement with core learning while retaining premium differentiation.

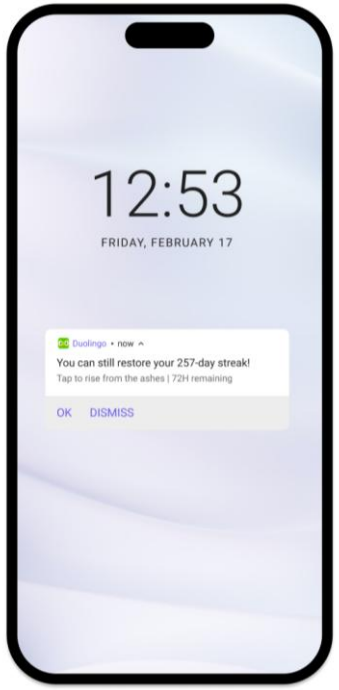
How it Works

1) Notification
Push sent at user's optimal engagement time

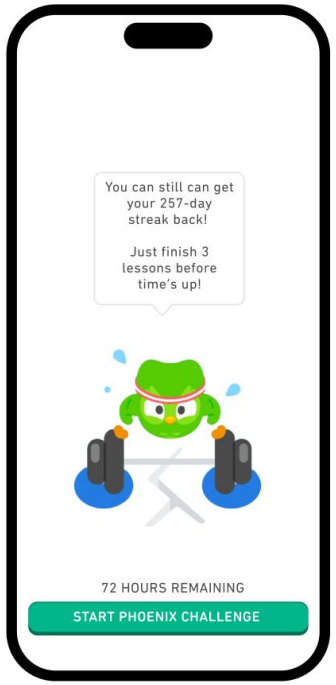
2) Opt-In
Full-screen modal presents challenge with explicit requirements

3) Active Challenge
Persistent progress tracker on home screen with 72-hour countdown

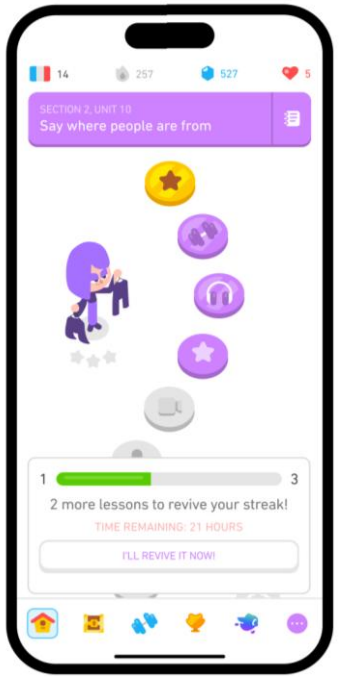
4) Celebration
Upon completion, streak is fully restored



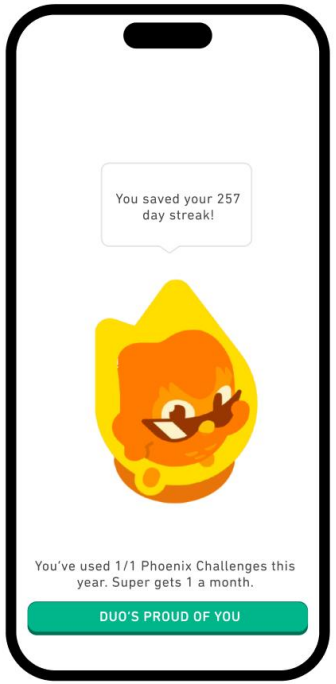
Notifications begin once criteria are met. Countdown creates urgency.



Full-screen takeover to capture attention. Visual progression from grey (loss) to orange (opportunity), then confirmation screen shown.



Persistent banner prevents forgetting. Dynamic countdown shifts from green (48+) to yellow (24+) to red (<24). Highlights next lesson to reduce friction.



Celebration rewards users for their effort. Visually, include animation of streak reviving. Usage counter creates scarcity awareness for soft upsell.

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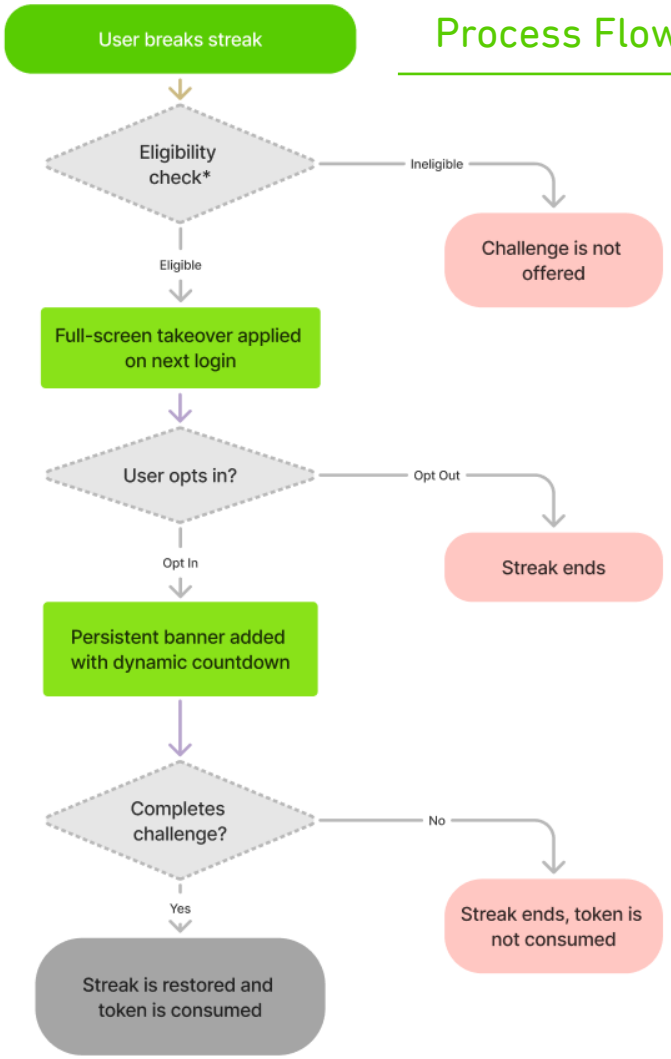
details and edge cases

Process Flow



*Users are eligible if they have a 100+ day streak, all streak freezes have been exhausted, and the Phoenix Challenge has not been completed recently (1/year free, 1/month Super & Max)

Integrating with Current Features

Streak Freeze: Phoenix Challenge is a secondary protection that does not replace Streak Freeze. To be eligible, they must have 0 freezes remaining.

Streak Repair: Phoenix Challenge replaces Streak Repair. By limiting to 1 a year, this creates a conversion opportunity to Super for users who repeat.

Super/Max: Super and Max users lose Streak Repair but can still repair streak monthly. This challenge encourages play and progression instead of passivity.

Post-Lesson Streak Counter: During the challenge, the post-lesson streak counter screen will be replaced 'X lessons until you revive your streak!" and will not count upwards.

Streak Society: Users in Streak Society retain membership during challenge period. They are only removed if the challenge fails.

XP/Quests: Wherever possible, the challenge should integrate into existing systems naturally and count towards requirements instead of creating parallel criteria.

Edge Cases

User Loses Streak During Challenge: User starts on Day 1 but misses Day 2. This is permissible. If they fulfill the 3 lesson criteria within the 72H window, it counts as success.

Timezone Confusion: The challenge timer will use absolute UTC timestamps, not local time. Displaying time remaining in hours and minutes prevents confusion.

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validation and testing

1 Problem Validation

First, we need to validate if the **problem actually exists**. I would analyze all users who lost streaks in the past 6 months, segmented by streak length (0-13, 14-49, 50-99, 100-364, 365+), and compare **iWAURR and D30 retention post-loss**.

The expected finding is that users with longer streaks (100-364, 365+) show statistically significant lower iWAURR and D30 retention, which would **validate that long-streak loss is more discouraging** (discounting the 0-13 group, which may be skewed by new users). However, if all streak lengths exhibit similar behaviour, the hypothesis fails.

2 Solution Validation

To validate if the Phoenix Challenge would **make users more willing to return**, I would pursue the following:

a) **User interviews** (n=30, lost 100+ day streaks) to understand why they quit and preferred challenge mechanics
b) **Preference survey** (n=500, active users with 100+ day streaks) to collect quantitative data

The expected outcome is that **users express heightened likeliness to quit upon losing a long streak** and interest in this comeback mechanism. If not, it does not address root motivation properly, and solution design will need to be changed.

3 A/B Testing

Via A/B testing on eligible users, we determine what implementation parameters **maximize engagement and completion**.

a) The control group will not be offered the Phoenix Challenge.
b) For treatment groups, we test **3 key parameters**: the time window (24/48/72h), the lesson requirement (2-6), and the enrolment model (opt-in vs auto-enrollment). These tests will show: what time spurs urgency but isn't overwhelming? What lesson requirement maximizes completion? Does opt-in create a feeling of ownership?

For each of these, we will measure primarily **measure completion rate** to determine which configuration is best. Overall, we will track our primary KPI **iWAURR** and **D7 retention** to see if there are material improvements to validate the feature. Key user actions to track include declining the challenge, failing the challenge, and lesson completes.

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expected impact and business case

40K
DAU Retained

$918K
Subscription Retention

$124K
Conversion Uplift

5.2x
Year 1 ROI

Calculations & Assumptions

DAU Retained: Starting with 8M users maintaining 365+ day streaks (a conservative estimate not including 100+ streaks), I assumed 2% of these streaks break each year and 50% of these users quit altogether. If the Phoenix Challenge can save 50% of these would-be churners, 40K users would be retained, with a nearly 1:1 impact on DAU.

Subscription Retention: Of these 40k users, I assumed 27% are Super subscribers (a 3x of baseline 9% due to these users being power users). At an annualized subscription price of $85, this amounts to $918K ARR.

Conversion Uplift: Of the remaining 73% of users saved, I assumed a conversion rate to Super at 5% upon seeing the value of the Phoenix Challenge monthly. Using the same annual subscription price, this results in $124K in additional ARR from previous non-subscribers.

Year 1 ROI: To calculate ROI, I estimated a dev cost of $200K, given that the feature is not technically complex. At this cost, the 1.04M annual benefit means a very fast 2-month payback period and a 5x ROI in year 1, with continuous benefits afterwards.

Risks & Mitigations

Devaluing Streaks: A vocal minority of long-streak users may complain that these comebacks "devalue" their unbroken streaks. While possible, this is a low-impact risk. If necessary, we can introduce exclusive badges for users who maintain unbroken streaks.

Pattern Abuse & Freeze Cannibalization: Users may deliberately use this to reduce need for Streak Freezes. This can reduce gem economy revenue but is largely mitigated via their continued subscription to Super. If necessary, we can escalate the difficulty of consecutive challenges.

Future Implementations

If this iteration of the Phoenix Challenge is successful, we can explore:

a) **Rise Again:** For users who miss this 72H window, their new streak is accelerated temporarily to assuage pain but not fully restored.
b) **Friendly Phoenix**: Users can spend their credit to help friends who lost their streak and quit (SURR). Their streak will be partially restored.

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